C1 Duminică, 20 februarie 2022

- Proiect
  - repartiții statistice
  - scheme clasice prob.
  - definiția clasică a probab
  - definiția axiomatică a prob
  - formula probab. totale
  - formula Bayes
  - LIVM - legi ale numerelor mari
  - TLC - teorema limită centrală
  - Selective din ce lucrăm - Exemple -
- Bibl < mat. cărți - date la prob (conțineau și stat. mat

1) Statistică descriptivă
2) Metode de estimare - în general
3) Metoda momentelor
4) Metoda verosimilității maxime
5) Intervale de încredere
6) Verificarea ipotezelor statistice
7) Teste de concordanță
   - χ^2 - Pearson
   - testul Kolmogorov-Smirnov

1) Statistică descriptivă

- date statistice
  - cu valori calitative < dairy, bus, etc ...
  - cu valori cantitative
- repartiția de frecvențe < frecv. absolută, frecv. relativă.
- reprezentări pt date statistice
  - cu dreptunghiuri
  - batoane, linie poligonală

- circulară
- cumulate ...

• indicatori de poziție pt date cantitative
caracteristica considerată – este reprezentată prin variabilă aleatoare asociată la X

• datele – selecție de volum – volume-dimensiune n
eșantion: $x_1, x_2, \dots, x_n$

• datele ordonate – statistică de ordine – pt date particulare – selecție – eșantion

$x_{(1)} \le x_{(2)} \le \dots \le x_{(n)}$.

$x_{(1)} = \min\{x_1, \dots, x_n\}$

$x_{(n)} = \max\{x_1, \dots, x_n\}$

• cuantile / cuartile

Q_2 = mediana = $Me = \begin{cases} x_{(k+1)}, & n = 2k+1 \\ \frac{x_{(k)} + x_{(k+1)}}{2}, & n = 2k \end{cases}$

x_1 x_2 x_3 x_4 x_5
2 5 7 3 4

$x_{(1)}$ $x_{(2)}$ $x_{(3)}$ $x_{(4)}$ $x_{(5)}$
2 3 4 5 7.

$Me = Q_2$; 50%

$n = 2k+1$
$n = 5$

2 4 7 5 9 2

2 2 4 5 7 9.

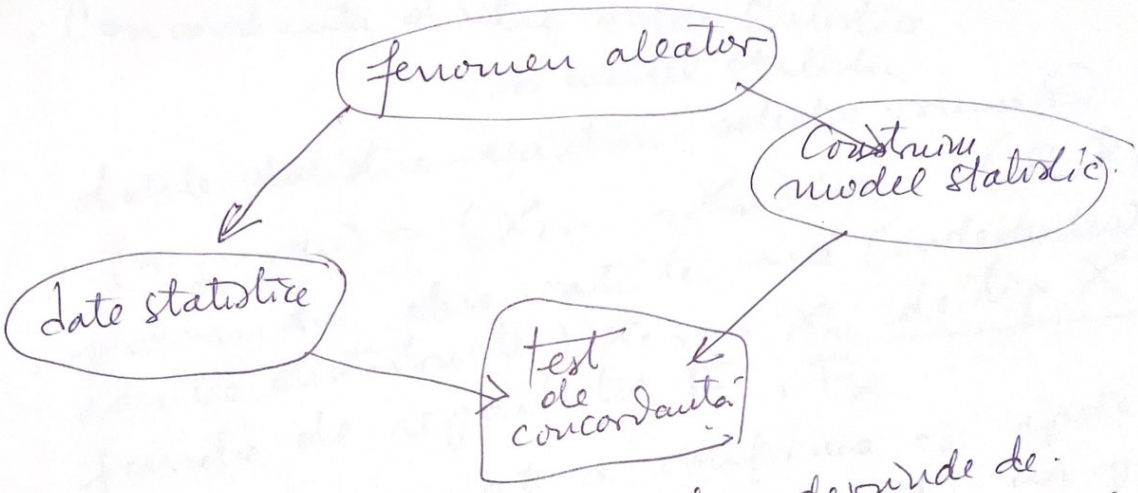
$n = 2k$
$n = 6$

$\frac{4+5}{2} = 4.5$

Q_1 este mediana pt datele; 25%.
↑ mai sfert.

$x_{(1)} \le \dots \le Me$

Q_3 = mediana pt $Me \le \dots \le x_{(n)}$ 75%
al treilea sfert



fenomenul aleator depinde de:
- natura lor
- de modul de colectare a datelor

(volumul datelor de selective) ← mic.
mare – suficient de mare – atunci avem repetate asimptotic

- cuantilă; $\alpha \in (0,1)$. q_α

$P(X < q_\alpha) \leq \alpha \leq P(X \leq q_\alpha)$

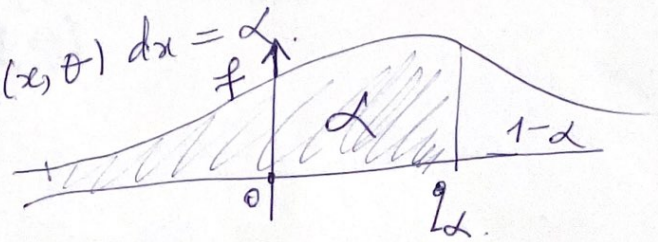
Atenție la caz discret!

(Caz absolut continuu: $P(X=a)=0$, $P(X \leq a) = P(X<a)$

q_α: $P(X \leq q_\alpha) = \alpha$ / $q_\alpha = F_X^{-1}(\alpha)$

$F_X(q_\alpha)$

$\int_{-\infty}^{q_\alpha} f(x,\theta)\,dx = \alpha$



$\int_{-\infty}^{\infty} f = 1$

• Concordanța dintre datele statistice și model statistic.

datele statistice - eșantion - selectiv volum (n)

$(x_1, \dots, x_n) = (X_1, \dots, X_n)(\omega)$ $(\Omega, \mathcal{K}, P)$ p

privim drd observațiile iid (independente și identic distribuite) $X_1, \dots, X_n$ de tip X de funcții de repartiție F, F_X

Problema: pt a confirma că ipoteza privind datele statistice $x_1, \dots, x_n$ ar provenii dintr-un model statistic procedăm astfel:

• Ce avem? ← (din situația reală) $x_1, \dots, x_n$.
Ce instrumente folosim?
• cum utilizăm acestea (datele, instrumentele)
• cum verificăm concluzia obținută.

Cele de mai sus sugerează la LAM - sugerează să comparăm fc de repartiție teoretică F cu o funcție construită din datele statistice $x_1, \dots, x_n$.

• Funcția de repartiție de selecție (empirică)

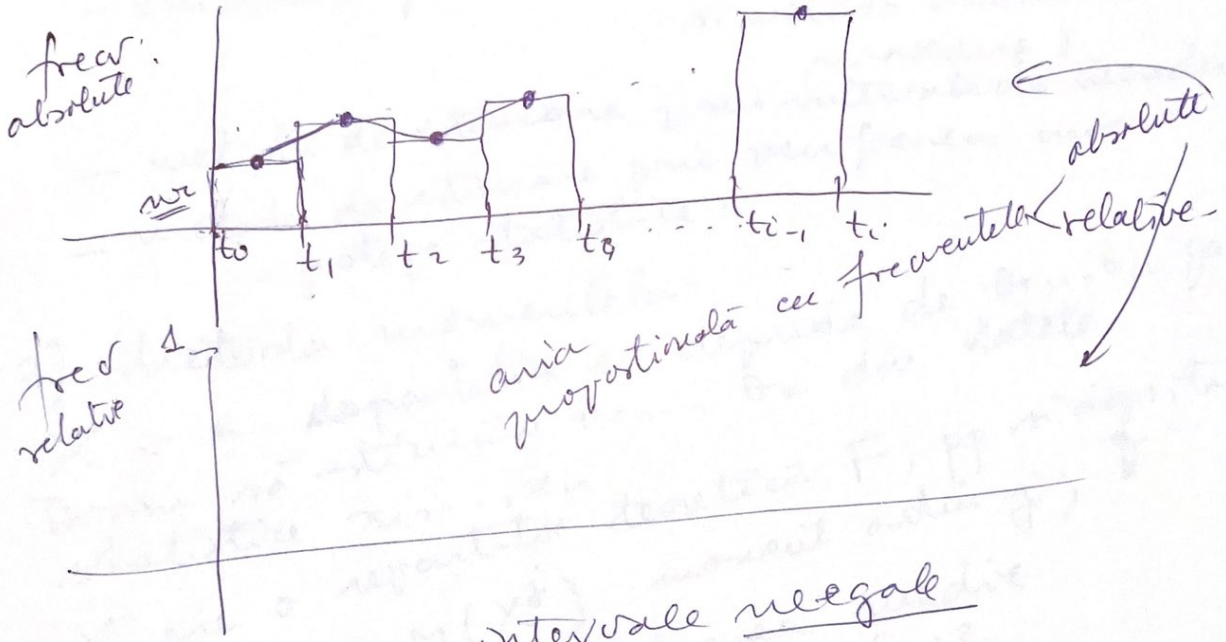
$F_n(y, \omega) = \frac{1}{n} \cdot \text{card}\{i / 1 \le i \le n, X_i(\omega) = x_i < y\}$

echivalent cu:

$F_n(y, \omega) = \frac{1}{n} \sum_{i=1}^{n} I_{\{X_i < y\}}(\omega)$

$I_{\{X_i < y\}}(\omega) = I_{\{X_i(\omega) < y\}} = \begin{cases} 1, & X_i(\omega) < y \\ 0, & \text{rest} \end{cases}$

• histograme, poligoni frecvențe; caz cumulativ



Dacă avem intervale neegale

dreptunghiurile au aria egală cu
frecvența absolută. γ_i = nr de $[t_{i-1}, t_i]$ în

intervale egale
n
relative pe $\frac{\gamma_i}{n}$

caz că sunt egale intervalele

$\frac{\gamma_i}{t_i - t_{i-1}}$ respectiv $\frac{\gamma_i}{n(t_i - t_{i-1})}$ în caz intervale neegale.

γ_i

2) a patra cumulat



mă conduce la grafic pt frecvențele empirice de selecție

1) grafic cumulată a acestei frecv cu o frecv teoretică

Tr 2

2) Metode de estimare - în general - (6)
- estimare punctuală (metoda momentelor și metoda verosimilității maxime)
- metoda de estimare prin intervale de încredere $q < p$
- metoda de estimare prin verificarea unor ipoteze statistice.

3) Metoda momentelor

X v.a. departiția ei depinde de $\theta_1, \dots, \theta_k$ par.
Dorim să estimăm $\theta_1, \dots, \theta_k$ din datele statistice $x_1, \dots, x_n$

X are o repartiție teoretică F, pp. are moment inițial de ordin j: $m_j = M(X^j)$

$m_1 = M(X) = m$ = valoarea medie

$D^2(X) = M(X^2) - (M(X))^2 = m_2 - (m_1)^2$

Din selecție calculăm $\overline{m}_j = \frac{1}{n}\sum_{i=1}^{n} (x_i)^j$ = moment de selecție de ordin j

Se consideră sistemul

$$M(X^j) = \overline{m}_j \quad , j = \overline{1,k}$$

$$m_j = \overline{m}_j, \quad j = \overline{1,k}$$

Exemplul 1 $X \sim N(m, \sigma^2)$

m, σ^2 necunoscute

$m = \theta_1$, $\sigma^2 = \theta_2$

$\begin{cases} m_1 = \overline{m}_1 \\ m_2 = \overline{m}_2 \end{cases}$

$\hat{\theta}_1 = \frac{1}{n}\sum_{i=1}^{n} x_i$

$\hat{\theta}_2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2 - \left(\frac{1}{n}\sum x_i\right)^2$

$\hat{\theta} = \hat{\theta}(x_1, \dots, x_n)$ estimație pt θ_1.

$\sigma^2 = M(X^2) - (M(X))^2 = \hat{\theta}_2 = \frac{1}{n}\sum x_i^2 - \left(\frac{1}{n}\sum x_i\right)^2$

Ex 2 $X \sim Be(p)$: $X : \begin{pmatrix} 1 & 0 \\ p & q=1-p \end{pmatrix}$

$x_1, \dots, x_n$

0 1 1 0 0 0 1 1 ...

Estimăm p: $\theta_1 = p$, $m_1 = \overline{m}_1$, $M(X) = 1 \cdot p + 0 \cdot q = p$

$p = m_1 = \overline{m}_1 = \frac{1}{n}\sum_{i=1}^{n} x_i$

$\hat{p} = \hat{p}(x_1, \dots, x_n) = \frac{1}{n}\sum_{i=1}^{n} x_i$, estimatie

Ex 3 $X \sim Bi(n; p)$

$X : \begin{pmatrix} 0 & 1 & 2 & \dots & k & \dots & n \\ q^n & - & - & - & C_n^k p^k q^{n-k} & \dots & p^n \end{pmatrix}$

dacă cunoaștem

$M(X) = np$

$D^2(X) = npq$ — teoretice

$D^2(X) = M(X^2) - [M(X)]^2$

$M(X^2) = npq + (np)^2$

dacă n = cunoscut, p = necunoscut $\Rightarrow$ $m_1 = \overline{m}_1 = \frac{1}{n}\sum_{i=1}^{n} x_i = \overline{x} = \overline{x}_n$

$m_1 = M(X) = np$

$np = \overline{x}_n$

$\hat{p} = \frac{1}{n}\overline{x}_n$

$\hat{p} = \hat{p}(x_1, \dots, x_n)$ estimatie pt p

dacă p necunoscut $\Rightarrow$

$m_1 = \overline{m}_1 \parallel \begin{cases} p = \frac{1}{n}\sum x_i \\ mpq + m^2p^2 = \frac{1}{n}\sum x_i^2 \end{cases}$

$m_2 = \overline{m}_2$

$M(X^2) = mpq + (mp)^2$

$\hat{p} = \overline{x}_n$

$m^2p^2 + mpq - \overline{m}_2 = 0$

$p = \frac{-mq \pm \sqrt{m^2q^2 + 4m^2\overline{m}_2}}{2m^2}$

$m_{1,2} = \frac{-\hat{p}\hat{q} + \sqrt{\hat{p}^2\hat{q}^2 + 4\overline{m}_2\hat{p}^2}}{2\hat{p}^2}$ $\quad \hat{q} = 1 - \hat{p}$

m întreg!

$\hat{m} = \left[\frac{-\hat{p}\hat{q} + \sqrt{\hat{p}^2\hat{q}^2 + 4\overline{m}_2\hat{p}^2}}{2\hat{p}^2}\right]$ — legea evenimentelor rare! / parte întreagă

Ex 4 repartiția Poisson

$X \sim Po(\lambda)$

Estimați pe λ din selecția 2 5 7 1 3 2 4

$x_1\ x_2\ x_3\ x_4\ x_5\ x_6\ x_7$

$X: \begin{pmatrix} 0 & 1 & 2 & \dots & k & \dots \\ e^{-\lambda} & \lambda e^{-\lambda} & \frac{\lambda^2}{2!}e^{-\lambda} & \dots & \frac{\lambda^k}{k!}e^{-\lambda} & \dots \end{pmatrix}$

$M(X) = D^2(X) = \lambda$, $\quad m_1 = \overline{m}_1 = \frac{1}{n}\sum x_i = \frac{1}{7}(2+5+7+1+3+2+4)$.

$M(X) = \lambda$

$\hat{\lambda} = \frac{24}{7}$

4) Metoda verosimilității maxime

Cazul absolut continuu:

$X \sim f(x;\theta)$, $\theta=(\theta_1,\dots,\theta_k)$

selecția: $x_1,\dots,x_n$

caz discret: $P(X_1=x_1,\dots,X_n=x_n)$. aleg θ astfel ca să fie maximă!

$\|$

$P(X_1=x_1)\cdots P(X_n=x_n)$.

revenim la cazul absolut continuu

$f(x_1;\theta),\dots,f(x_n;\theta)$. Știind că selecția este i.i.d

Construim funcția de verosimilitate:

$L(x_1,\dots,x_n;\theta_1,\dots,\theta_k)=f(x_1;\theta)\cdots f(x_n;\theta)$

Alegem $\theta_1,\dots,\theta_k$ care maximizează

$L(x_1,\dots,x_n;\theta_1,\dots,\theta_k)$

Avem pb: $\max_{\theta_1,\dots,\theta_k} L(x_1,\dots,x_n;\theta_1,\dots,\theta_k)$

Se determină pct staționare

$\frac{\partial L}{\partial \theta_j}=0$, $j=\overline{1,k}$

Mai simplu: $\max_{\theta_1,\dots,\theta_k} \ln L(x_1,\dots,x_n;\theta_1,\dots,\theta_k)$

$l(x_1,\dots,x_n;\theta_1,\dots,\theta_k)=\ln L(x_1,\dots,x_n;\theta_1,\dots,\theta_k)$

$$\left\{\frac{\partial \ln L}{\partial \theta_j}=0,\ j=\overline{1,k}\right\}$$

Studiem dacă $\exists$ max!
Trebuie ca hessiana lui $\ln L$ să fie negativ definită

Ce înseamnă $\ln L$?

$$\ln L = \ln\left(f(x_1,\theta)\cdots f(x_n;\theta)\right) = \sum_{i=1}^{n} \ln f(x_i;\theta)$$

$$\frac{\partial \ln L}{\partial \theta_j} = \sum_{i=1}^{n} \frac{\partial \ln f(x_i;\theta)}{\partial \theta_j} = 0 \quad , j=\overline{1,k}$$

Exemple

Ex 1 Cazul exponențial: $\exp(\lambda)$, $\lambda>0$

$$f(x;\theta) = \begin{cases} \lambda e^{-\lambda x} & , x>0 \\ 0 & , x\le 0 \end{cases} \quad , \lambda>0$$

λ = param. Se dă selecția: $x_1,\dots,x_n$
Să se estimăm pe λ pe baza selecției $x_1,\dots,x_n$.

Sol Construim fc. de verosimilitate

$$L(x_1,\dots,x_n;\lambda) = f(x_1;\lambda)\cdot f(x_2;\lambda)\cdot\dots\cdot f(x_n;\lambda)$$

$$= \lambda e^{-\lambda x_1}\cdot \lambda e^{-\lambda x_2}\cdot\dots\cdot \lambda e^{-\lambda x_n} =$$

$$= \lambda^n e^{-\lambda x_1}\cdot e^{-\lambda x_2}\cdots e^{-\lambda x_n} =$$

$$= \lambda^n e^{-\lambda x_1 - \lambda x_2 \cdots - \lambda x_n} =$$

$$= \lambda^n e^{-\lambda(x_1+\dots+x_n)} =$$

$$= \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}$$

$$= \lambda^n \cdot e^{-\lambda \sum_{i=1}^{n} x_i}$$

$$L(x_1,\dots,x_n;\lambda) = \lambda^n \cdot e^{-\lambda \sum_{i=1}^{n} x_i}$$

$$\ln L(x_1,\dots,x_n;\lambda) = \ln\left(\lambda^n e^{-\lambda\sum_1^n x_i}\right) =$$

$$= \ln\lambda^n + \ln e^{-\lambda\sum_1^n x_i} = n\ln\lambda - \lambda\sum_1^n x_i$$

$$\boxed{\ln L(x_1,\dots,x_n;\lambda) = n\ln\lambda - \lambda\sum_{i=1}^n x_i}$$

$$\frac{\partial \ln L}{\partial \lambda} = n\cdot\frac{1}{\lambda} - \sum_{i=1}^n x_i = 0.$$

$$\lambda = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\frac{1}{n}\sum_1^n x_i} = \frac{1}{\overline{x}_n}.$$

$$\boxed{\hat{\lambda} = \frac{1}{\overline{x}_n}} \qquad \hat{\lambda} = \hat{\lambda}(x_1,\dots,x_n).$$

este pct max?

$$\left.\begin{array}{l} \frac{\partial^2 \ln L}{\partial \lambda^2} = -\frac{n}{\lambda^2} < 0 \end{array}\right\} \Rightarrow \hat{\lambda} \text{ este pct de } \underline{\underline{\text{max}}}$$

$\Rightarrow \hat{\lambda}$ = estimator pt param λ.

Ex 2 $X \sim \mathcal{N}(m, \sigma^2)$ m, σ^2 param.

$x_1,\dots,x_n$ selectie $\in \mathbb{R}$

$f(x; m, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$, $x \in \mathbb{R}$.

Să estimăm $\theta_1 = m$, $\theta_2 = \sigma^2$ pe baza selecției $x_1,\dots,x_n$.

Soluție Scriem funcția de verosimilitate

$L(x_1,\dots,x_n; m, \sigma^2) = f(x_1; m, \sigma^2)\cdot\dots\cdot f(x_n; m, \sigma^2)$.

$$L(x_1,\dots,x_n;m,\sigma^2)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x_1-m)^2}{2\sigma^2}}\cdot\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x_2-m)^2}{2\sigma^2}}\cdots\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(x_n-m)^2}{2\sigma^2}}$$

$$=\frac{1}{\sigma^n(\sqrt{2\pi})^n}e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-m)^2}$$

$$\ln L=\ln\frac{1}{\sigma^n(\sqrt{2\pi})^n}e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-m)^2}$$

$$=-\ln\sigma^n(\sqrt{2\pi})^n+\ln e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-m)^2}\cdot\ln e$$

$$\ln L=-\ln\sigma^n-\ln(\sqrt{2\pi})^n-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-m)^2\cdot 1$$

$$\ln L=-n\ln\sigma-n\ln(\sqrt{2\pi})-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-m)^2$$

$$\frac{\partial\ln L}{\partial m}=0$$

$$\frac{\partial\ln L}{\partial\sigma^2}=0$$

$\theta_1=m$

$\theta_2=\sigma^2$

$$\ln L=-\frac{n}{2}\ln\theta_2-n\ln\sqrt{2\pi}-\frac{1}{2\theta_2}\sum(x_i-\theta_1)^2$$

$$\frac{\partial\ln L}{\partial\theta_1}=0$$

$$\frac{\partial\ln L}{\partial\theta_2}=0$$

$$\frac{\partial\ln L}{\partial\theta_1}=+\frac{1}{2\theta_2}\sum_{i=1}^{n}2(x_i-\theta_1)=0$$

$$\frac{\partial\ln L}{\partial\theta_2}=-\frac{n}{2}\cdot\frac{1}{\theta_2}+\frac{1}{2}\frac{1}{(\theta_2)^2}\sum(x_i-\theta_1)^2=0$$

$$\begin{cases} \sum_{i=1}^{n}(x_i-\theta_1)=0 \\ -n+\frac{1}{\theta_2}\sum_{i=1}^{n}(x_i-\theta_1)^2=0 \end{cases} \Bigg| \begin{array}{l} \sum_{i=1}^{n} x_i - n\theta_1 = 0 \\ \theta_2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\theta_1)^2 \end{array}$$

$$\hat{\theta}_1 = \frac{1}{n}\sum_{i=1}^{n} x_i \quad \Rightarrow \boxed{\hat{\theta}_1 = \bar{x}_n} \qquad \boxed{\hat{m} = \bar{x}_n}$$

$$\hat{\theta}_2 = \hat{\sigma}^2 = \frac{1}{n}\sum (x_i - \hat{\theta}_1)^2 = \frac{1}{n}\sum (x_i - \bar{x}_n)^2$$

$$\boxed{\hat{\sigma}^2 = \frac{1}{n}\sum (x_i - \bar{x}_n)^2}$$

Este pct de maxim?

Hessiana lui $\ln L$

$$\begin{pmatrix} \frac{\partial^2 \ln L}{\partial \theta_1^2} & \frac{\partial^2 \ln L}{\partial \theta_1 \partial \theta_2} \\ \frac{\partial^2 \ln L}{\partial \theta_1 \partial \theta_2} & \frac{\partial^2 \ln L}{\partial \theta_2^2} \end{pmatrix} \Bigg|_{\substack{m=\hat{m} \\ \sigma^2=\hat{\sigma}^2}}$$

$$\begin{cases} \frac{\partial \ln L}{\partial \theta_1} = \frac{1}{\theta_2}\sum_{i=1}^{n}(x_i-\theta_1) = \frac{1}{\theta_2}\left(\sum_{i=1}^{n} x_i - n\theta_1\right) \\ \frac{\partial \ln L}{\partial \theta_2} = -\frac{n}{2\theta_2} + \frac{1}{2(\theta_2)^2}\sum_{i=1}^{n}(x_i-\theta_1)^2 \end{cases}$$

$$\frac{\partial^2 \ln L}{\partial \theta_1^2} = -\frac{n}{\theta_2} \qquad \frac{\partial^2 \ln L}{\partial \theta_2 \partial \theta_1} = -\frac{1}{\theta_2^2}\sum_{i=1}^{n}(x_i-\theta_1)$$

$$\left(\frac{1}{x^2}\right)' = (x^{-2})' = -2x^{-3}$$

$$\frac{\partial^2 \ln L}{\partial \theta_2^2} = \frac{n}{2(\theta_2)^2} - \frac{2}{2(\theta_2)^3} \sum_{i=1}^{n} (x_i - \theta_1)^2$$

$$\text{Hess}(\ln L) = \begin{pmatrix} -\frac{n}{\theta_2} & -\frac{1}{\theta_2^2}\sum(x_i-\theta_1) \\ -\frac{1}{\theta_2^2}\sum(x_i-\theta_1) & \frac{n}{2(\theta_2)^2} - \frac{1}{\theta_2^3}\sum(x_i-\theta_1)^2 \end{pmatrix} \Big|_{\substack{\theta_1=\hat{\theta}_1 \\ \theta_2=\hat{\theta}_2}}$$

$-\frac{n}{\theta_2} < 0$ ok!

$$\left|\text{Hess}\ln L\Big|_{\substack{\theta_1=\hat{\theta}_1 \\ \theta_2=\hat{\theta}_2}}\right| > 0 \quad -\left(\text{Hess}\ln L\right)\Big|_{\substack{\theta_1=\hat{\theta}_1 \\ \theta_2=\hat{\theta}_2}} =$$

$$= \begin{pmatrix} -\frac{n}{\theta_2} & 0 \\ 0 & \frac{n}{2\theta_2^2} \end{pmatrix}$$

$$\sum x_i - n\hat{\theta}_1 = \sum(x_i - \hat{\theta}_1)$$

$$\sum x_i - n\bar{x}_n = \sum x_i - n\cdot\frac{1}{n}\sum x_i = 0$$

$$\frac{n}{2\theta_2^2} - \frac{1}{\theta_2^3}\sum(x_i-\theta_1)^2 = \frac{1}{\theta_2^2}\left[\frac{n}{2} - \frac{1}{\theta_2}\sum(x_i-\theta_1)^2\Big|_{\theta_2=\hat{\theta}_2}\right]$$

$$\sum(x_i-\theta_1)^2 = n$$

$$= \frac{1}{\theta_2^2}\left[\frac{n}{2} - n\right] = \frac{1}{\theta_2^2}\left(-\frac{n}{2}\right) = \frac{-n}{2\theta_2^2}$$

Gata!

$$\det \text{Hess}\ln L\Big|_{\substack{\theta_1=\hat{\theta}_1 \\ \theta_2=\hat{\theta}_2}} = \frac{n^2}{2\theta_2^3} > 0$$

Ex 3 Cazul Poisson(λ), $\lambda > 0$

$$f(x;\lambda) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x = 0, 1, 2, \dots$$

$$X : \begin{pmatrix} 0 & 1 & 2 & \dots & x & \dots \\ - & - & - & - & \frac{\lambda^x}{x!} e^{-\lambda} & \dots \end{pmatrix}$$

estimam pe λ pe baza unei selectii $x_1, \dots, x_n \in \{0, 1, 2, \dots\}$

Solutie Scriem functia de verosimilitate

$$L(x_1, \dots, x_n; \lambda) = f(x_1, \lambda) \cdot \dots \cdot f(x_n, \lambda) =$$

$$= \frac{\lambda^{x_1}}{x_1!} e^{-\lambda} \cdot \frac{\lambda^{x_2}}{x_2!} e^{-\lambda} \cdot \dots \cdot \frac{\lambda^{x_n}}{x_n!} e^{-\lambda} =$$

$$= \frac{\lambda^{x_1} \cdot \lambda^{x_2} \cdot \dots \cdot \lambda^{x_n}}{x_1! x_2! \dots x_n!} \underbrace{e^{-\lambda} \cdot e^{-\lambda} \dots e^{-\lambda}}_{n \text{ ori}} =$$

$$= \frac{\lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} e^{-n\lambda}$$

$$\ln L(x_1, \dots, x_n; \lambda) = \ln\left(\frac{\lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n (x_i!)} e^{-n\lambda} \right) =$$

$$= \ln \lambda^{\sum_{i=1}^n x_i} - \ln \prod_{i=1}^n (x_i!) + \ln e^{-n\lambda}$$

$$= \left(\sum_{i=1}^n x_i \right) \ln \lambda - \ln \prod_{i=1}^n (x_i!) - n\lambda \cdot \underbrace{\ln e}_{=1}$$

$$\ln L = \left(\sum_{i=1}^n x_i \right) \ln \lambda - \ln \prod_{i=1}^n (x_i!) - n\lambda.$$

$$\frac{\partial \ln L}{\partial \lambda} = \frac{d \ln L}{d\lambda} = \left(\sum_{i=1}^n x_i \cdot \frac{1}{\lambda} - n \right) = 0$$

$\hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} x_i$, $\boxed{\hat{\lambda} = \bar{x}_n}$

Este pct de maxim?

$\frac{\partial^2 \ln L}{\partial \lambda^2} = \frac{\partial}{\partial \lambda}\left[\left(\sum x_i\right)\cdot\frac{1}{\lambda} - n\right] =$

$= -\frac{1}{\lambda^2}\sum_{i=1}^{n} x_i < 0 \Rightarrow$ este pct de max $\hat{\lambda}$ pt L.

Ex Caz binomial $Bi(m,p)$

$X : \begin{pmatrix} 0 & 1 & 2 & \dots & x & \dots & m \\ q^m & - & - & - & C_m^x p^x q^{m-x} & \dots & p^m \end{pmatrix}$

$q = 1-p$

$f(x; m, p) = C_m^x p^x (1-p)^{m-x}$

$x_1, \dots, x_n \in \{0, 1, \dots, m\}$

$\boxed{\text{nu e demonstrat}} = P(X = x)$

Presupunem m cunoscut. Estimăm probab p pe baza unei selecții $x_1, \dots, x_n$.

Soluție Scriem fc. de verosimilitate

$L(x_1, \dots, x_n; p) = P(X = x_1) \cdot \dots \cdot P(X = x_n) =$

$$= f(x_1; n, p) \cdot \dots \cdot f(x_n; n, p) =$$

$$= C_n^{x_1} p^{x_1} (1-p)^{n-x_1} \cdot \dots \cdot C_n^{x_n} p^{x_n} (1-p)^{n-x_n} =$$

$$= \left(C_n^{x_1} \cdot C_n^{x_2} \cdots C_n^{x_n}\right) \cdot p^{\sum_1^n x_i} \cdot (1-p)^{nn - \sum_1^n x_i}$$

$$\ln L = \ln\left(\prod_{i=1}^n C_n^{x_i}\right) + \ln p^{\sum_1^n x_i} + \ln(1-p)^{nn - \sum_1^n x_i}$$

$$\ln L = \ln\left(\prod_1^n C_n^{x_i}\right) + \left(\sum_1^n x_i\right) \ln p + \left(nn - \sum_1^n x_i\right) \ln(1-p)$$

$$\frac{\partial \ln L}{\partial p} = \left(\sum_1^n x_i\right) \cdot \frac{1}{p} - \left(nn - \sum_1^n x_i\right) \frac{1}{1-p} = 0.$$

$$(1-p) \sum_1^n x_i - p\left(nn - \sum x_i\right) = 0,$$

$$\sum_1^n x_i - p \sum_1^n x_i - p \cdot nn + p \sum_1^n x_i = 0$$

$$pnn = \sum x_i \quad ; \quad p = \frac{1}{n} \cdot \frac{1}{n} \sum x_i = \bar{x}_n.$$

$$\boxed{\hat{p} = \frac{\bar{x}_n}{n}} \text{ estimate of } p$$

the point max?

$$\frac{\partial^2 \ln L}{\partial p^2} = -\left(\sum x_i\right) \cdot \frac{1}{p^2} - \left(nn - \sum x_i\right) \frac{1}{(1-p)^2}$$

$$\left.\frac{\partial^2 \ln L}{\partial p^2}\right|_{p=\hat{p}} = -\frac{\hat{p} nn}{\hat{p}^2} - (nn - \hat{p} nn) \cdot \frac{1}{(1-\hat{p})^2} =$$

$$= -\frac{nm}{\hat{p}} - nm(1-\hat{p})\cdot\frac{1}{(1-\hat{p})^2}$$

$$= -\frac{nm}{\hat{p}} - \frac{nm}{1-\hat{p}} < 0$$

$+$ $+$ $\Rightarrow$ este pct de max!

Ex 5: Cazul Weibull cu trei parametri

α, β, γ

$$f(x;\alpha,\beta,\gamma) = \frac{\beta}{\alpha}\left(\frac{x-\gamma}{\alpha}\right)^{\beta-1}\cdot \exp\left[-\left(\frac{x-\gamma}{\alpha}\right)^{\beta}\right]$$

$\gamma \le x < +\infty$ $\quad \alpha,\beta,\gamma$ cu $\alpha>0$, $\beta>0$, $\gamma\in\mathbb{R}$

α = param. de scală

β = caracteristică a formei

γ = param poziție.

Schemă estimare α,β,γ:

selective $x_1,\dots,x_n \ge \gamma$

$$L(x_1,\dots,x_n;\alpha,\beta,\gamma) = f(x_1,\alpha,\beta,\gamma)\cdot\dots\cdot f(x_n;\alpha,\beta,\gamma)$$

$$= \left(\frac{\beta}{\alpha}\right)^n \prod_{i=1}^{n}\left(\frac{x_i-\gamma}{\alpha}\right)^{\beta-1}\cdot \exp\left[-\sum_{i=1}^{n}\left(\frac{x_i-\gamma}{\alpha}\right)^{\beta}\right]$$

$$\ln L = \ln\left(\frac{\beta}{\alpha}\right)^n + \ln\prod_{i=1}^{n}\left(\frac{x_i-\gamma}{\alpha}\right)^{\beta-1} + \ln\exp\left[-\sum_{1}^{n}\left(\frac{x_i-\gamma}{\alpha}\right)^{\beta}\right]$$

$$\ln L = n\ln\beta - n\ln\alpha + (\beta-1)\sum_{i=1}^{n}\ln\frac{x_i-\gamma}{\alpha} - \sum\left(\frac{x_i-\gamma}{\alpha}\right)^{\beta}$$

Puncte staționare

$$\frac{\partial \ln L}{\partial \alpha} = 0$$

$$\frac{\partial \ln L}{\partial \beta} = 0$$

$$\frac{\partial \ln L}{\partial \gamma} = 0$$

$$\ln L = n \ln \beta - n \ln \alpha + (\beta - 1) \sum_{i=1}^{n} \left[\ln (x_i - \gamma) - \ln \alpha \right] - \sum \left(\frac{x_i - \gamma}{\alpha} \right)^{\beta}$$

$$\ln L = n \ln \beta - n \ln \alpha + (\beta - 1) \sum \ln (x_i - \gamma) - n(\beta - 1) \ln \alpha - \sum_{1}^{n} \left(\frac{x_i - \gamma}{\alpha} \right)^{\beta}$$

$$\frac{\partial \ln L}{\partial \alpha} = -\frac{n}{\alpha} - \frac{n(\beta - 1)}{\alpha} + \beta \sum_{i=1}^{n} \left(\frac{x_i - \gamma}{\alpha} \right)^{\beta - 1} \frac{x_i - \gamma}{\alpha^2} = 0$$

$$-\frac{n}{\alpha} - \frac{n(\beta - 1)}{\alpha} + \frac{\beta}{\alpha^{\beta + 1}} \sum_{1}^{n} (x_i - \gamma)^{\beta} = 0$$

$$\frac{\partial \ln L}{\partial \beta} = \frac{n}{\beta} + \sum_{1}^{n} \ln (x_i - \gamma) - n \ln \alpha - \sum_{1}^{n} \left(\frac{x_i - \gamma}{\alpha} \right)^{\beta} \ln \frac{x_i - \gamma}{\alpha} = 0$$

$$\frac{\partial \ln L}{\partial \gamma} = -(\beta - 1) \sum_{i=1}^{n} \frac{1}{x_i - \gamma} + \frac{\beta}{\alpha} \left(\frac{x_i - \gamma}{\alpha} \right)^{\beta - 1} = 0$$

...

Ex 6 Cazul rep geometrice

$f(x,\rho) = (1-\rho)\rho^x \qquad \rho \in (0,1)$

$x = 0, 1, 2, \dots$

$\| \; P(X=x)$

$x_1, \dots x_n$

$L(x_1, x_i; \rho) = P(X=x_1) \cdots P(X=x_n) =$

$= f(x_1,\rho) \cdots f(x_n,\rho) =$

$= (1-\rho)\rho^{x_1} \cdot (1-\rho)\rho^{x_2} \cdots (1-\rho)\rho^{x_n} =$

$= (1-\rho)^n \cdot \rho^{\sum_1^n x_i}$

$L(x_1, \dots x_n; \rho) = (1-\rho)^n \rho^{\sum_1^n x_i}$

$\ln L = \ln(1-\rho)^n \cdot \rho^{\sum_1^n x_i}$

$\ln L = \ln(1-\rho)^n + \ln \rho^{\sum_1^n x_i}$

$\ln L = n \ln(1-\rho) + \sum_1^n x_i \cdot \ln \rho$

$\frac{\partial \ln L}{\partial \rho} = \frac{-n}{1-\rho} + \sum_1^n x_i \cdot \frac{1}{\rho} = 0$

$-n\rho = (1-\rho)\sum x_i$

$-n\rho = \sum x_i - \rho \sum x_i$

$\rho(\sum x_i - n) = \sum x_i$

$\rho = \frac{\sum x_i}{\sum x_i - n} = \frac{\frac{1}{n}\sum x_i}{\frac{1}{n}\sum x_i - 1} = \frac{\bar{x}_n}{\bar{x}_n - 1}.$

$$\boxed{\hat{\rho} = \frac{\bar{x}_n}{\bar{x}_n - 1}}$$

Este max?

$$\frac{\partial^2 \ln L}{\partial \rho^2} = -\frac{n}{(1-\rho)^2} - \sum_1^n x_i \cdot \frac{1}{\rho^2} < 0$$

$\Rightarrow \hat{\rho}$ este pct max pt L.

Ex 7 Rep. gamma

$X \sim \Gamma(\alpha, \beta)$ sau $\gamma(\alpha, \beta)$, $\alpha, \beta > 0$.

$$\begin{cases} M(X) = \alpha\beta \\ M(X^2) = \alpha(\alpha+1)\beta \end{cases}$$

$$D^2(X) = M(X^2) - (M(X))^2 = \alpha^2\beta + \alpha\beta - \alpha^2\beta^2$$

Estimare cu metoda momentelor.

$$\left.\begin{aligned} m_1 &= \overline{m}_1 \\ m_2 &= \overline{m}_2 \end{aligned}\right\} \quad \begin{aligned} M(X) &= \alpha\beta = \frac{1}{n}\sum_1^n x_i \\ M(X^2) &= \alpha(\alpha+1)\beta = \frac{1}{n}\sum_1^n x_i^2 \end{aligned}$$

$$\alpha\beta = \bar{x}_n$$

$$\alpha^2\beta + \alpha\beta = \frac{1}{n}\sum_1^n x_i^2$$

$$\alpha \cdot \alpha\beta + \alpha\beta = \frac{1}{n}\sum_1^n x_i^2$$

$$(\alpha+1)\cdot \bar{x}_n = \frac{1}{n}\sum_1^n x_i^2$$

$$\alpha + 1 = \frac{\sum x_i^2}{\sum x_i}$$

$$\hat{\beta} = \frac{\sum x_i^2 - \sum x_i}{\sum x_i}$$

$$\hat{\alpha} = \frac{\sum x_i^2 - \sum x_i}{\sum x_i}$$